CO543/CO5430 Computer Vision Project Proposal

Stereo Disparity and Depth Estimation from Rectified Stereo Images

Group	G03	**Project ID**	P12
Area	Stereo Depth	**Status**	Topic selected and approved
Members	E/23/127: H.M.K.I. Herath E/23/188: K.M.M.Y. Kumarasinge E/23/343: S.B.N.S. Samarawickrama E/23/347: S.D.M.P. Sandanayake	**Course**	CO543/CO5430 Computer Vision
Dataset	Middlebury Stereo 2021 scenes	**Dataset Link**	https://vision.middlebury.edu/stereo/data/scenes2021/
Repository	https://github.com/cepdnaclk/e23-co5430-Stereo-disparity_depth-estimation	**Main Methods**	Block Matching, StereoSGBM, post-processing

1. Project Remark and Problem Statement

This project builds a computer vision system that estimates a disparity/depth map from a rectified stereo image pair. Stereo depth estimation matches pixels between two viewpoints to compute disparity, larger disparity means closer objects, and smaller disparity means farther objects. The aim is to generate accurate disparity maps and evaluate them against ground-truth data.

2. Motivation

Depth perception is important in robotics, autonomous navigation, 3D scene understanding, obstacle detection, and augmented reality. Unlike monocular depth estimation, stereo depth uses geometric information from two views, making it a strong Computer Vision topic with clear algorithmic steps, measurable metrics, and meaningful visual outputs.

3. Related Work

- Block Matching finds disparity by comparing small windows between the left and right images. It's simple and fast, but struggles with texture less areas, repeated patterns, and occlusions.
- SGBM improves on this by adding smoothness constraints across multiple directions, giving cleaner results while still being practical to implement.
- Middlebury and KITTI are standard datasets for testing stereo algorithms, using ground-truth disparity maps and metrics like RMSE and bad-pixel rate.

4. Dataset and Preprocessing

The primary dataset is the Middlebury Stereo 2021 scenes, with rectified stereo pairs and ground-truth disparity maps. A small KITTI subset may be added later as an optional generalization test. Dataset files won't be committed to GitHub; download and setup steps will be documented in the README.

Preprocessing Step	Purpose
Load stereo pair	Read left/right rectified images and corresponding ground-truth disparity.
Resize/crop if needed	Keep computation realistic while preserving important scene details.
Grayscale conversion	Prepare input for BM/SGBM while optionally testing color input effects.
Disparity scaling/masking	Handle invalid pixels and normalize disparity for visualization/evaluation.
Result split	Use selected scenes for parameter tuning and held-out scenes for final testing.

5. Proposed Method

Baseline: The baseline method will use OpenCV StereoBM or a basic block matching approach. Important parameters such as block size, number of disparities, pre-filtering, and texture threshold will be tested. This provides a simple reference result for comparison.

Improved Method: The improved method will use OpenCV StereoSGBM with parameter tuning and optional post-processing such as speckle filtering, median filtering, weighted least squares filtering, and left-right consistency checking. The final output will be a dense disparity map and an optional relative depth visualization.

6. Minimum, Expected, and Stretch Goals

Goal Level	Planned Achievement
Minimum Goal	Load Middlebury stereo pairs, implement BM baseline, generate disparity maps, and compute at least RMSE/bad-pixel rate.
Expected Goal	Implement and tune SGBM, compare with BM, and present visual + quantitative improvements.
Stretch Goal	Add post-processing, parameter ablation, optional KITTI test, or relative depth conversion using calibration information where available.

7. Evaluation Plan

Evaluation will use task-appropriate stereo/depth metrics: RMSE, bad-pixel rate, absolute relative error where applicable, and qualitative disparity/depth maps. The report will compare BM and SGBM using the same scenes, include parameter studies, and discuss failure cases such as occlusion boundaries, reflective surfaces, weak texture, repeated patterns, and depth discontinuities.

8. Timeline

Period	Task	Deliverable
Week 1	Finalize dataset download, repository structure, and reading of Middlebury format.	Dataset proof, README update
Week 2	Implement preprocessing and BM baseline.	Baseline disparity maps and initial metrics
Week 3-4	Implement SGBM and tune major parameters.	Improved disparity maps and comparison table
Week 5	Run ablation/parameter study and optional post-processing.	Metric tables, plots, and qualitative results
Week 6	Analyze failures, prepare demo script/notebook, clean repository.	Failure cases and demo evidence
Final Week	Write final report and prepare presentation.	Report, slides, final GitHub repository

9. Risks and Mitigation

Risk	Mitigation
Wrong disparity scaling or invalid-pixel handling.	Verify dataset format early and test evaluation code on known sample outputs.
Poor quality disparity maps from default parameters.	Use systematic parameter tuning for block size, numDisparities, smoothness penalties, and filtering.
High-resolution images may be computationally heavy.	Use selected scenes, resizing/cropping, and runtime reporting.
Difficult areas such as occlusions and textureless regions.	Report these as failure cases and compare BM vs SGBM behavior clearly.

10. Expected Deliverables

- GitHub repository with README, setup instructions, dataset instructions, source code, notebooks/scripts, and example outputs.
- BM baseline and SGBM improved method with quantitative comparison and qualitative disparity maps.
- Parameter study showing how block size, disparity range, smoothness, and filtering affect results.
- Final IEEE-style report, contribution table, presentation slides, and working inference/demo script.

11. References

[1] G. Pan, T. Sun, T. Weed, and D. Scharstein, "2021 Mobile Stereo Datasets with Ground Truth," Middlebury Stereo Datasets, 2021. [Online]. Available: https://vision.middlebury.edu/stereo/data/scenes2021/
[2] OpenCV Documentation, OpenCV 4.13.0, 2025. [Online]. Available: https://docs.opencv.org/4.13.0/

12. AI Tool Use Declaration

AI tools may be used for writing support, debugging, explanation, and documentation improvement. All generated code and text will be reviewed, tested, and understood by the group before submission.